

CS 402: Distributed Systems & Cloud Architecture

Department of Computer Science | Synapse University | Fall 2026

4 Credit Hours

Lecture & Lab

Prereq: CS 301

Instructor Information

Instructor: Prof. Elena Rostova
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Office: Turing Hall, Room 412
Office Hours: Tue / Thu 2:00 PM – 4:00 PM
Website: synapse.edu/cs402

Course Logistics

Lectures: Mon / Wed 10:00 AM – 11:30 AM
Location: Meridian Science Center 104
Lab Session: Friday 1:00 PM – 3:00 PM
Lab Location: Apex Computing Lab 2
Teaching Asst: Marcus Vance
(m.vance@synapse.edu)

Course Overview & Objectives

CS 402 provides an in-depth study of distributed algorithms, cloud infrastructure design, and scalable system architecture. Students will explore fundamental principles governing consensus, fault tolerance, replication, and distributed storage. Hands-on laboratory assignments will focus on building robust services using modern microservices design patterns.

Upon successful completion of this course, students will be able to:

1. Analyze and apply classic distributed consensus protocols such as Raft and Paxos.
2. Architect fault-tolerant storage engines leveraging consistent hashing and event-driven messaging.
3. Evaluate performance trade-offs under CAP theorem constraints and eventual consistency models.
4. Deploy and monitor containerized microservices across cloud clusters using modern orchestrators.

Prerequisites & Course Materials

- ✓ **Required Text:** *Designing Scalable Distributed Systems*, 3rd Ed., Rostova & Vance, Vertex Academic Press, 2024.
- ✓ **Recommended Reference:** *Cloud Architecture Guidelines*, Meridian Press, 2025.
- ✓ **Software Tools:** Docker, Kubernetes local cluster (minikube/k3s), Go programming environment (v1.22+), Git.